

## 10 POWER TIPS FOR INTEL-BASED SYSTEMS

**The right installation will pave the way for trouble-free operation and let you harness the full potential of your PC. Keep in mind the following tips while buying and installing an Intel-based processor**

**1 Keeping your cool**  
Since processors are continuously stressing on faster speeds, cooling becomes an important issue—gift your processor a rated heatsink and cooling fan combination. Most processors come with the heatsink and cooling fan in tow, but you may require a better cooling solution, especially in the case of the Pentium 4 processors.

**2 How much thermal paste?**  
If your heatsink is covered by thermal tape, you don't need to apply any thermal paste. Otherwise, apply

just a dab—too much thermal paste and it will ooze out. Also make sure there are no air bubbles.

**3 Power to run**  
Due to additional power requirements, all Pentium 4 processors need the extra +12 V rail to run correctly. Therefore, make sure the power supply for your new processor has the necessary four-pin +12 V connector that goes into the ATX12V connector on your motherboard—both your power supply and motherboard should support this connector.

**4 Go for quality**  
Don't skimp on the motherboard—a high-quality motherboard may make you a few thousand rupees poorer, but will guarantee far greater stability and support for hardware, chipset and BIOS. You'll also get more features such as more overclocking options and integrated peripherals.

**5 BIOS updates**  
Make sure you buy a motherboard that offers regular BIOS and driver updates on the company Web site. This spells great support for newer hardware and helps

iron out the bugs and flaws that may crop up. Newer BIOS updates can also mean significant performance boosts.

**6 To HyperThread or not?**  
Intel's new 3.06 GHz processor now incorporates a technology called HyperThreading, which enables the processor to perform far better when running multiple applications simultaneously. However, this comes at a price and you should opt for it only if you need the high performance for executing multiple tasks. You will also need to confirm if your motherboard supports HyperThreading.

**7 A question of memory**  
The current scenario gives users the choice of SDRAM, DDR and RDRAM memory in a Pentium 4-based system. DDR memory offers high data transfer bandwidth and is priced lower than RDRAM. However, RDRAM's higher data transfer rate makes it ideally suited to demanding applications such as gaming, CAD modelling and graphics-intensive, texture-heavy applications. Also, RDRAM needs to be installed in pairs, unlike the DDR or SDRAM—this needs to be kept in mind while buying memory modules. These three types of memories cannot be used on the same system, therefore, choose your camp with care as you will not be able to change your memory type later without also changing the motherboard.

**8 Willamette or Northwood**  
While the newer Northwood Pentium 4 processors built around the 0.13-micron fabrication process are more popular than their older 0.18-micron Willamette counterparts, the latter is still available in the market. The Willamette processors draw more power and therefore generate more heat. Also, the Northwood processors integrate twice the amount of L2 cache (512 KB) as compared to 256 KB integrated by the Willamettes.

**9 Special drivers**  
Intel supplies drivers for the new 800-series chipsets (including the 845, 845-DDR and 850 chipsets) called the Intel Application Accelerator driver—another fancy name for the UDMA drivers. These increase the performance of your IDE devices quite significantly. They offer better boot times and also surpass the 137 GB limitation of the IDE drives in the native Widows drivers.

**10 Application support for special instruction sets**  
Intel processors have special instruction sets such as SSE2 that help deliver better performance with certain types of data. Ensure that you look out for applications that support these special instruction sets when using a P4 processor. In some cases, you might need to install specific patches that let these applications use the special instruction sets.



The Pentium 4's Thermal Heat Spreader protects and dissipates heat from its core